

		Department of Computer Science & Engineering (CSE) School of Engineering, Technology & Sciences INDEPENDENT UNIVERSITY, BANGLADESH (IUB)		
Computer Science and Engineering (CSE) Program (<i>Undergraduate</i>)				
MINI Project Report		CSE 303 Lab		Summer 2026
Course Code	CSE303	Course Title	Database Management	
Course Type	Core	Co-offered (if any)	CSE303, CSC401, CEN401	
Credit Value	4 (T: 3, L: 1)	Contact Hours / Week	3 hours (theory) 2 hours (Lab)	
Year	3 rd Year	Semester	7 th Semester	
Name: Mr. XYZ		ID: 22025214	Sec: 1-(3)	
CLO4	Apply cutting edge DBMS tools and techniques to implement solutions to a complex database problem with rational justification.		PLO-e: Modern tool usage: Select/apply modern engineering & IT tools (prediction/modeling) appropriately and understand their limitations.	
Tasks/Topics covered and Skill Measured: - Designing the database (ER-Diagram with detail specifications). - Create and Manage the Database using MySQL. - User Access control. - Application of Database Normalization Techniques. - Scenario based Database Recovery Mechanism. - Application of Hashing and Indexing. - Database Connectivity and front-end design.				
Acronyms and Legends used: MA – Marks Allotted MO – Marks Obtained				
MA - 150		MO- Out of 150 - Out of 15 -		
Project Definition - Hotel Management Database System				
Designing a Hotel Management Database A hotel chain wants to develop a Hotel Management System to manage hotels, rooms, guests, employees, reservations, services, and billing records.				

The hotel chain may have multiple branches. Each hotel branch has a unique hotel ID, hotel name, address, city, contact number, and star rating. Rooms belong to hotels. Each room has a unique room ID, room number, room type, floor number, capacity, nightly rate, and availability status.

Guests register before booking rooms. Each guest has a unique guest ID, full name, phone number, email, address, nationality, and identification number. Guests can make reservations. Each reservation includes reservation ID, booking date, check-in date, check-out date, reservation status, and number of guests.

The hotel employs staff members. Each employee has employee ID, full name, designation, salary, joining date, and employment status. Guests may use additional services such as laundry, transportation, spa, and room service. Service records store service type, date, quantity, and charge.

Bills are generated for room stays and services. Each bill contains bill ID, total amount, taxes, discounts, payment method, and payment status. The system should answer questions such as: room availability, upcoming check-ins, hotel occupancy rates, highest-spending guests, and monthly revenue.

Task -1	Designing the database	MA - 20	MO --	
Identify and list all entities, attributes for each entity, domain of each attribute, candidate keys and primary key for each entity. - Each row should be used for one entity. - Add/remove rows as appropriate.				
Entity name	Attribute list	Domain of each attribute	Candidate key(s) for each entity	Primary key for the entity
MA - 2	MA - 2	MA – 3	MA – 2	MA - 1
MO -	MO -	MO -	MO -	MO -
Draw the detailed ER diagram for the identified entity set and associated information. Your design must include: entity sets with all attributes, attribute types and domain, keys, relationships with cardinality, relationship sets (with attributes), weak entity sets, roles in relationships. All appropriate symbols and notations must be used to prepare the diagram.				
MA (Completeness) - 3	MA (Accuracy) - 3		MA (VIVA) - 4	
MO -	MO -		MO -	

Task -2	Create, Manage and Access Control of the Database using MySQL.	MA - 35	MO --
Create all the Tables, Relation between tables with all the attributes (domain), integrity constrains and relations based on the above ER Diagram.			
MA (Completeness) - 2		MA (Accuracy) - 1	MA (VIVA) - 2
MO -		MO -	MO -
Write comprehensive SQL Queries to perform the followings: a. Show guest name, phone number, and nationality for all guests. b. Display employee name, designation, and salary for all employees. c. Find guests whose names contain "Rana" and show their phone numbers. d. Show reservation ID, guest name, room number, and check-in date for all reservations. e. Find employees whose salary is greater than the average salary of all employees. f. Display the total number of rooms available in each hotel branch. g. Find hotel branches where the average room rate is greater than 5,000. h. Show room number, room type, and nightly rate for all rooms. i. Display guest name and check-in date for all active reservations. j. Find the total bill amount generated by each guest.			
MA (Completeness) - 5	MA (Accuracy) - 5		MA (VIVA) - 5
MO -	MO -		MO -

Write comprehensive SQL triggers to perform the following:			
a. Automatically check room availability before inserting a reservation. If a room is already unavailable, prevent the reservation. Write a sample INSERT statement showing how the trigger works. b. Recalculate the final bill amount before updating a bill record. The bill amount is calculated as follows: $Final_Amount = Total_Amount + Taxes - Discounts$ The trigger should automatically update the Final_Amount whenever the bill information is modified. c. Store deleted reservation information in a reservation log table. Whenever a reservation record is deleted, the trigger should save all relevant reservation details along with the deletion date and time into the log table.			
MA (Completeness) - 5	MA (Accuracy) - 5		MA (VIVA) - 5
MO -	MO -		MO -
Task -3	Scenario based Database Recovery Mechanism.	MA - 18	MO -
The hotel database backup system uses RAID Level 4 to improve data reliability and ensure continuous access to hotel information. In RAID Level 4, data is divided into fixed-size blocks and stored across multiple data disks using block-level striping. A separate dedicated parity disk stores the parity block. The parity block is calculated using the XOR operation on all data blocks. In this system, there are: 6 data disks: D1, D2, D3, D4, D5, D6 1 dedicated parity disk: P - Each disk block stores 4 bits - Parity is calculated using XOR operation The following hospital database blocks are stored in the RAID 4 system:			
Data Block	Description	4-bit Binary Value	Stored in
Block 1 (D1)	Hotel Data	1010	D1
Block 2 (D2)	Room Data	1100	D2
Block 3 (D3)	Guest Data	0111	D3
Block 4 (D4)	Employee Data	1001	D4

Block 5 (D5)	Reservation Data	0011	D5
Block 6 (D6)	Billing Data	1110	D6
The parity block is stored in the dedicated parity disk P. Now perform the following: - Draw the RAID Level 4 structure. - Show the detailed calculation of the parity block (P). - Assume Data Block D5 fails. Show the detailed calculation to recover the lost data. - Verify that the recovered value matches the original Reservation Data block. - Briefly explain why RAID Level 4 can recover data when a single disk fails.			
MA (Completeness) - 6			MA (VIVA) - 12
MO -			MO -

Task -4	Database Normalization.	MA - 22	MO -
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Give an Unnormalized sample database table for the Hotel Database system.

Guest ID	Guest Name	Room ID	Room Type	Hotel Name	Employee Name	Services Used	Reservation ID	Bill Amount
G001	Rahim Ahmed	R101	Deluxe	Grand Hotel	Hasan Ali	Laundry, Spa	RES001	15000
G002	Karim Hasan	R102	Standard	City Inn	Nabila Islam	Room Service, Transport	RES002	12000
G003	Nila Sultana	R103	Suite	Grand Hotel	Imran Hossain	Spa, Laundry	RES003	25000
G004	Arif Khan	R104	Standard	Sea View Hotel	Tania Akter	Laundry, Room Service	RES004	10000
G005	Nusrat Jahan	R105	Deluxe	Royal Palace Hotel	Shakil Ahmed	Spa, Transport	RES005	18000
G006	Fahim Hossain	R106	Executive Suite	City Inn	Rakib Hasan	Room Service, Laundry	RES006	22000

The parity block is stored in the dedicated parity disk P.

Now perform the following:

a. Apply 1NF to normalize the given table and produce the resultant table.

b. Apply 2NF to normalize the table(s) produced in 1NF and present the resultant tables.

c. Apply 3NF to normalize the table(s) produced in 2NF and present the resultant tables.

MA (Completeness) - 6	Ma (Correctness) - 6	MA (VIVA) - 10
MO -	MO-	MO -

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Task -5	Hashing and Indexing.	MA - 25	MO -
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Scenario

The Hotel Management Database System maintains a **ROOM** table containing information about hotel rooms. The database administrator wants to create an index on **Room_ID** so that room records can be searched efficiently.

ROOM Table

Room_ID	Room_Number	Room_Type
101	101A	Standard
102	102A	Deluxe
103	103A	Suite
104	104B	Standard
105	105B	Deluxe
106	106B	Suite
107	107C	Standard
108	108C	Deluxe
109	109C	Suite
110	110D	Standard
111	111D	Deluxe
112	112D	Suite
113	113E	Presidential

The index key is: **Patient_ID**

The insertion sequence is: 109, 102, 111, 105, 101, 113, 107, 104, 110, 103, 112, 106, 108

Assume:

- Every node can contain at most 3 keys.
- The trees must remain balanced.

Now perform the following:

- Apply the B-Tree algorithm to create a B-Tree for the ROOM table.
- Apply the B+ Tree algorithm to create a B+ Tree for the ROOM table.
- Demonstrate how the system searches for **Room_ID** = 104 using the B+Tree.
- Demonstrate how the system deletes **Room_ID** = 112 from the B-Tree

MA (Completeness) - 6	Ma (Correctness) - 6	MA (VIVA) - 13
MO -	MO-	MO -

Task -6	User Authentication and Database Connectivity.	MA - 30	MO -
Develop a simple Hotel Management Web Application with user authentication and database connectivity. The system should provide: 1. User Registration: New users can create an account using a username, email, and password. 2. User Login: Registered users can log in securely using their credentials. 3. Room Management Page: After successful login, users can access a page to manage room records. 4. The page should support basic CRUD operations: • Create: Add a new room record. • Read: View all room records. • Update: Edit existing room information. • Delete: Remove a room record from the database. The application must connect to a database to store user accounts and room information permanently.			
MA (Completeness) - 6		MA (VIVA) - 12	
MO -		MO -	

